

R

LV.0 → LV.1 · AI ENGINEER

13 R-words that move you from **Level 0** → 1.

RAG, RLHF, ReLU, RoPE & more. **Plain-English** definitions, one swipe each, plus the R-named tools and minds to follow.



↓ UP NEXT • TERMS

The R stack.

RAG

RLHF

ReLU

RoPE

RL

RNN

Reasoning

ReAct

Reranker

RAGAS

R^2 score

Replication

RDBMS

R

RECURRENT NEURAL NETWORK • FOUNDATIONS

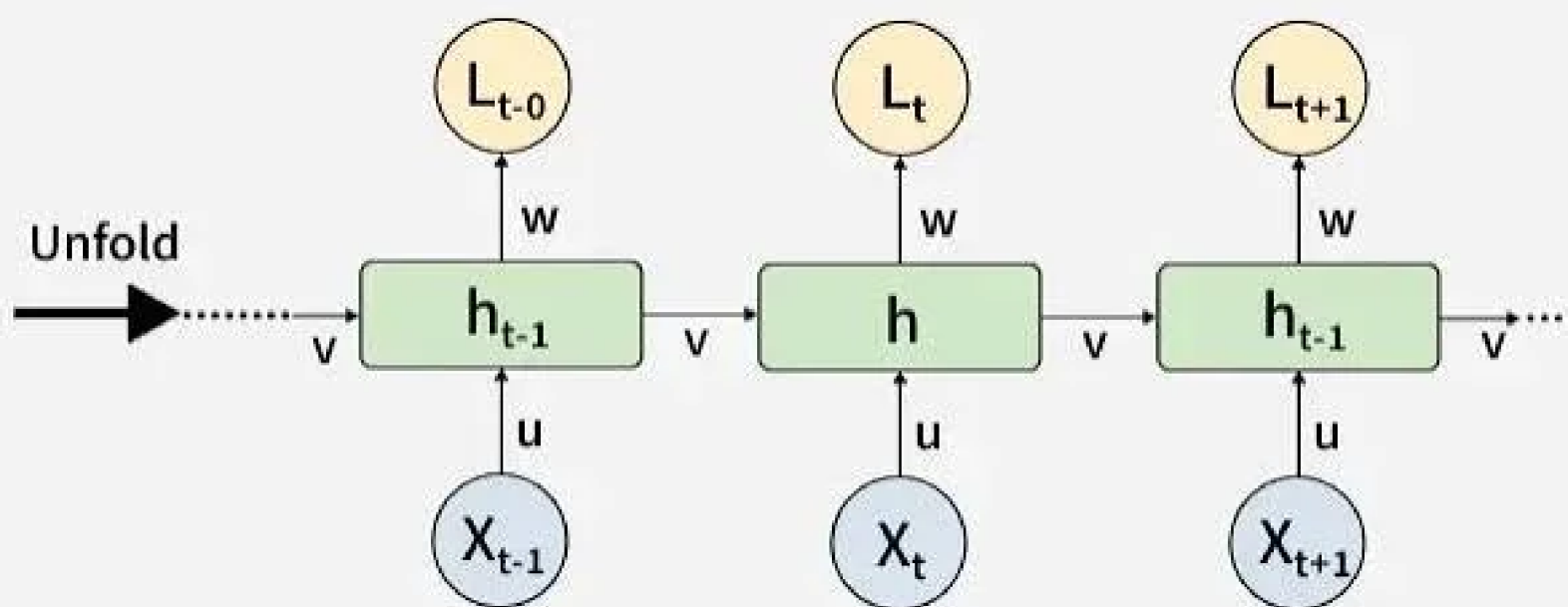
RNN

hidden state • carried across time

A network that processes sequences by **passing a hidden state between time steps**, the standard for sequence modeling before Transformers took over.

USED IN

- Sequence modeling
- Speech
- Time series



UNROLLED STEPS • MEMORY LOOP

RECTIFIED LINEAR UNIT • ACTIVATIONS

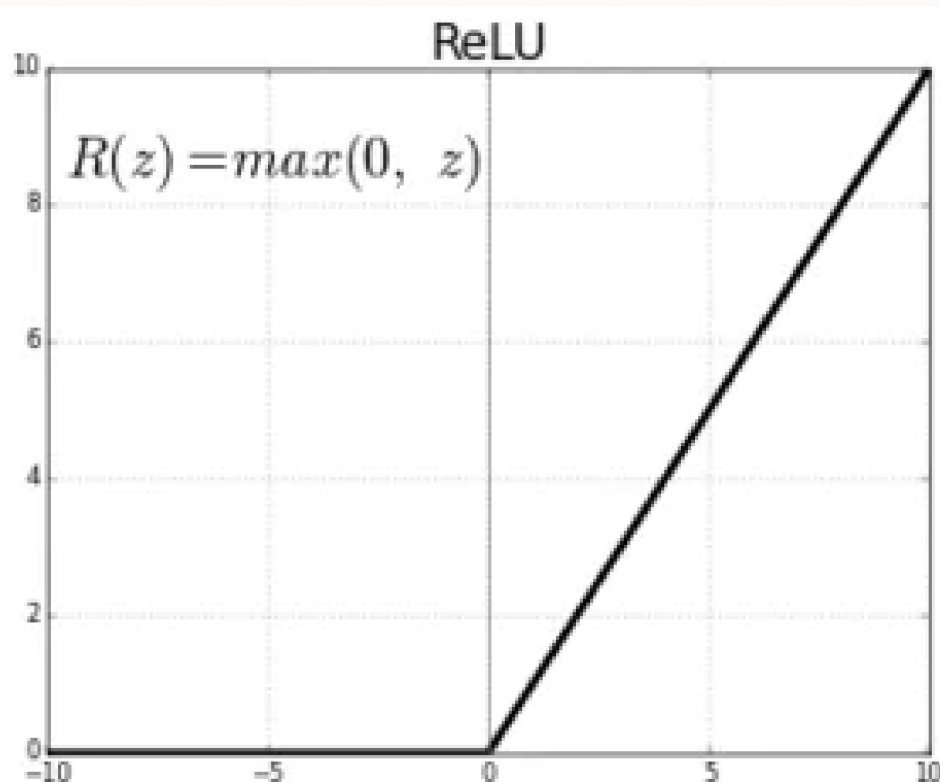
ReLU

positive in • positive out • else zero

An activation that passes positive inputs through and zeroes the rest, **simple, fast, and the default** in most deep networks for avoiding vanishing gradients.

USED IN

- Deep networks
- CNNs
- Hidden layers



$$F(X) = \max(0, X)$$

REINFORCEMENT LEARNING • LEARNING

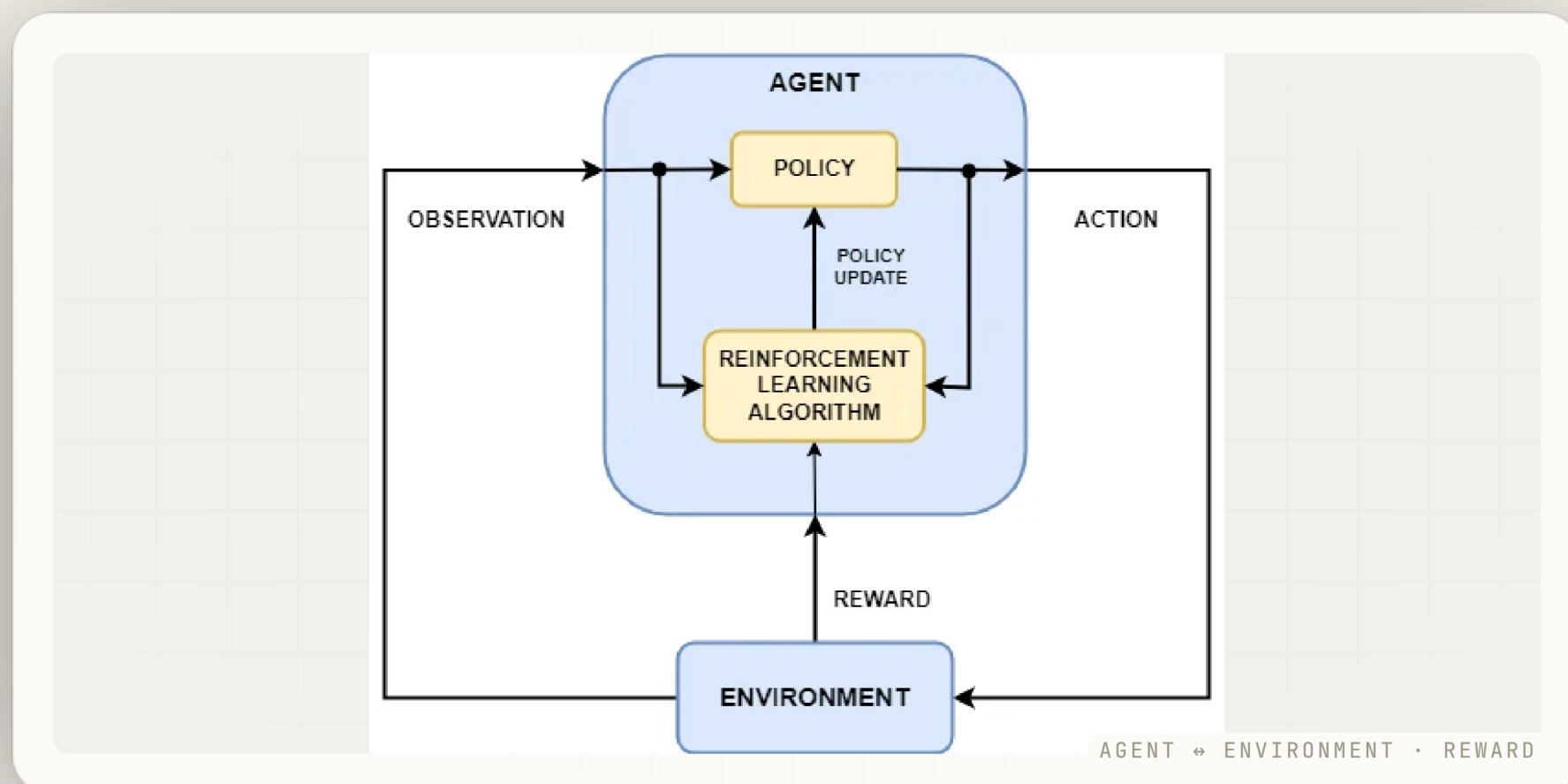
RL

trial and error • maximize reward

A paradigm where an agent learns by **taking actions to maximize cumulative reward** through trial and error, powering RLHF, robotics, and game-playing AI.

USED IN

- RLHF
- Robotics
- Game-playing AI



RL FROM HUMAN FEEDBACK • ALIGNMENT

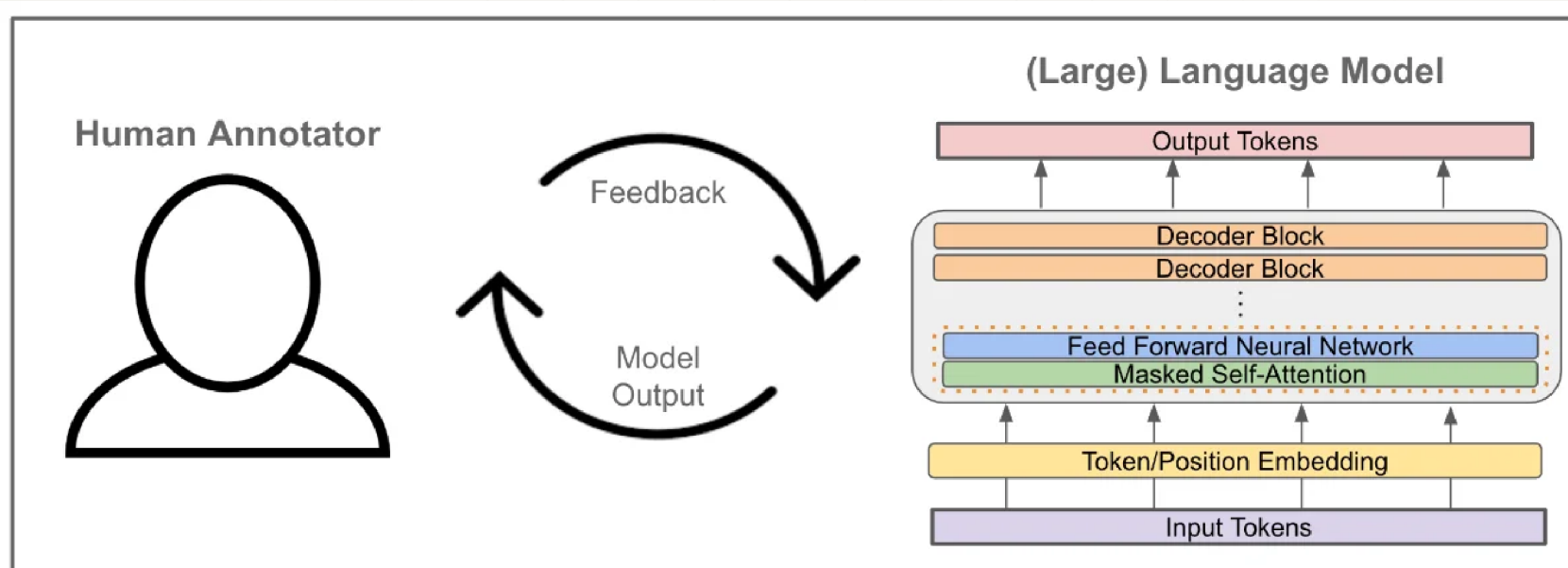
RLHF

human ranks • reward model • align

Human preference rankings train a reward model that then guides an LLM via RL, **making outputs more helpful and aligned** with what people actually want.

USED IN

- LLM alignment
- Chat assistants
- Safety tuning



RANKINGS → REWARD → PPO

REASONING • INFERENCE

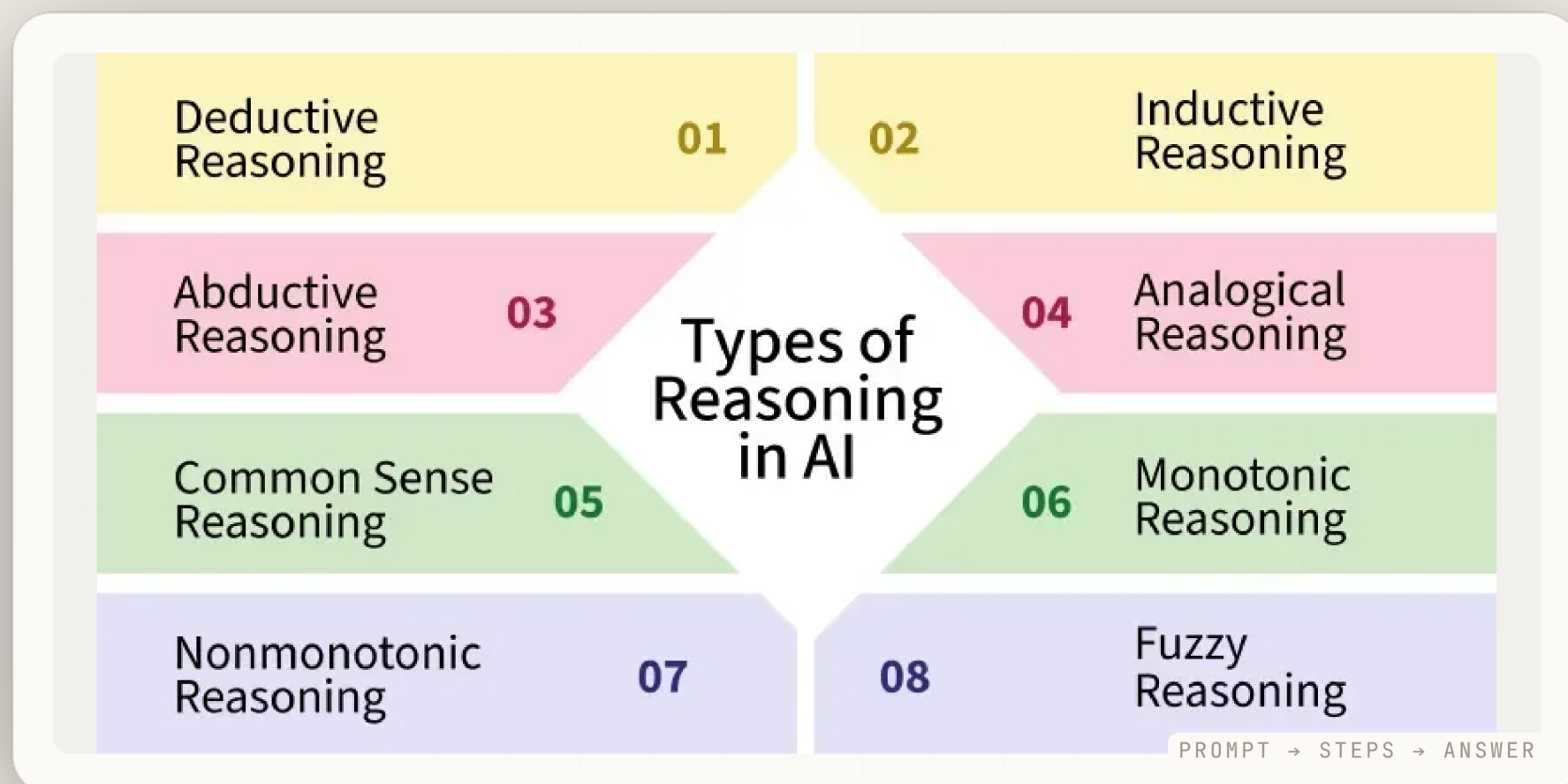
Reasoning

multi-step • chain of thought

A model's ability to **work through multi-step logical or mathematical problems**, often improved via chain-of-thought prompting or dedicated reasoning training.

USED IN

- Math & code
- Chain-of-thought
- Agents



REASON + ACT • AGENTS

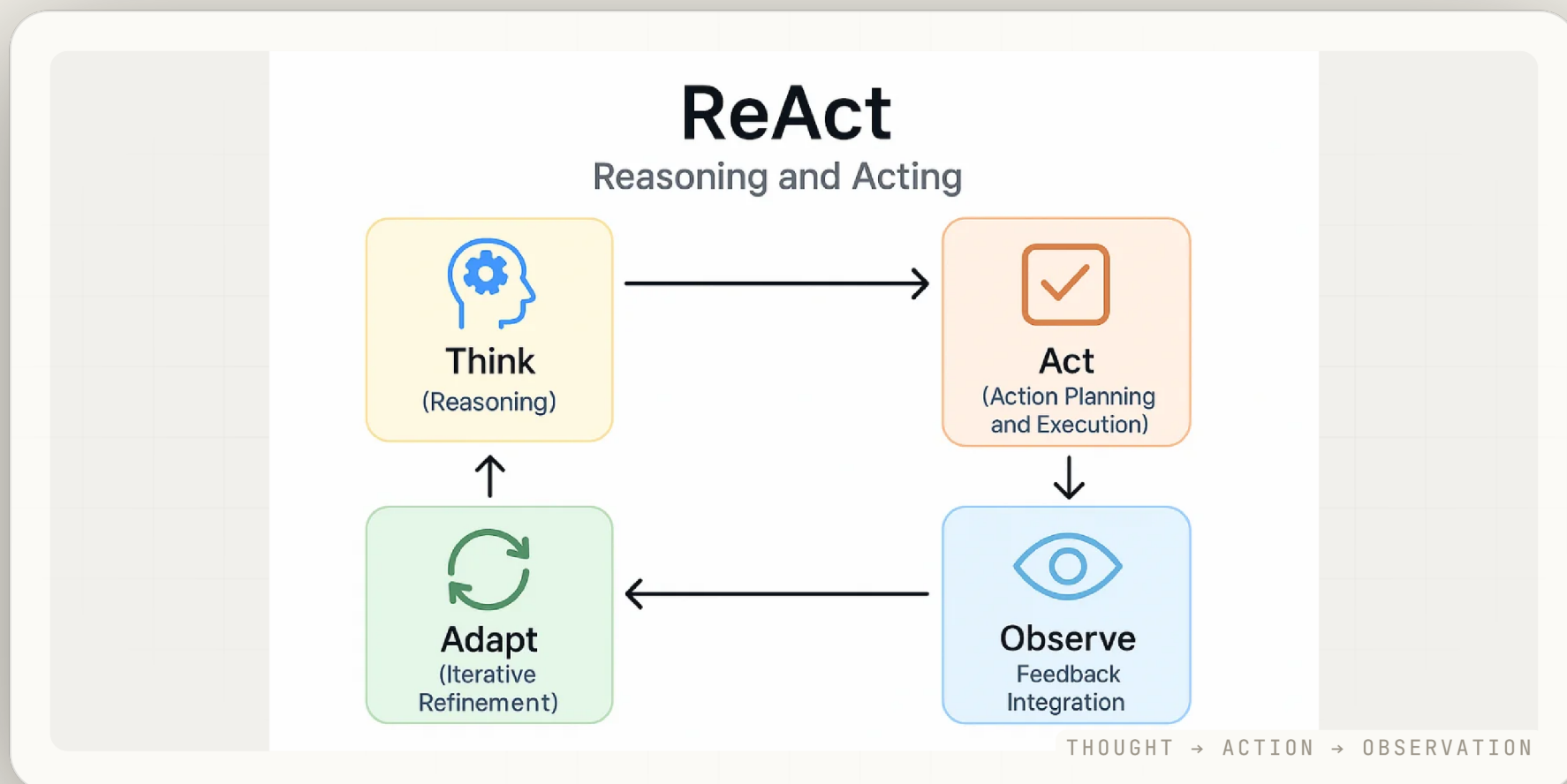
ReAct

think • then act • then observe

A framework where an LLM **alternates between reasoning and taking actions** like tool calls, the backbone of many modern agent architectures.

USED IN

- AI agents
- Tool use
- Multi-step tasks



RETRIEVAL-AUGMENTED GENERATION • RETRIEVAL

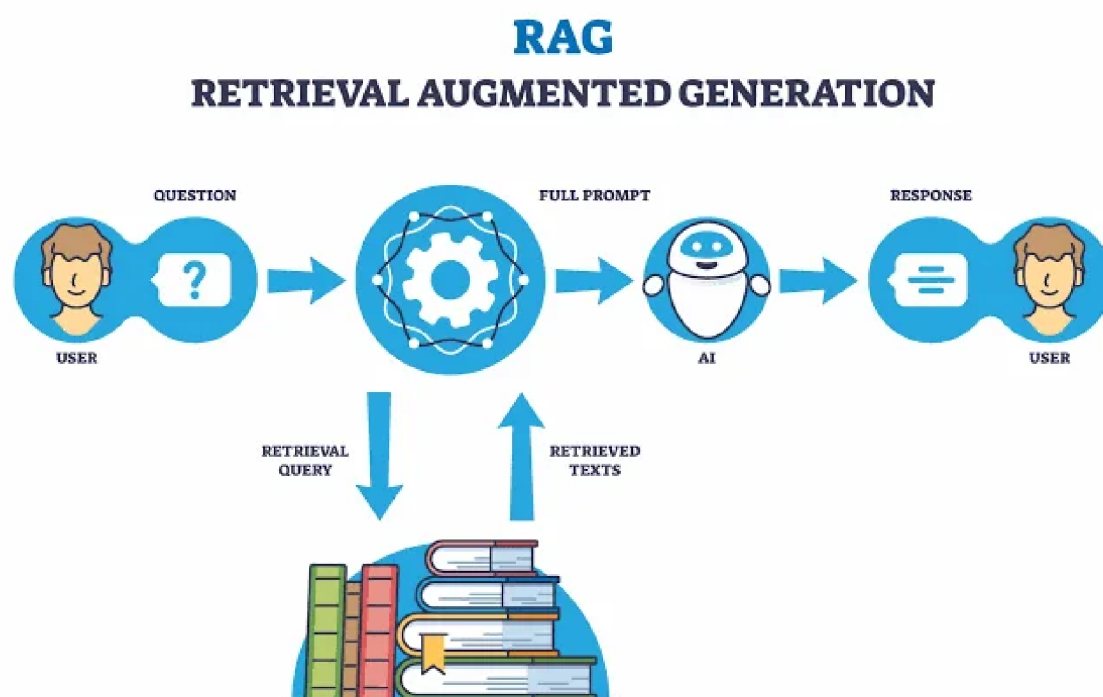
RAG

retrieve • augment • generate

An architecture that retrieves relevant documents and feeds them into an LLM's context before generation, **improving factual accuracy and cutting hallucination.**

USED IN

- Q&A over docs
- Chat assistants
- Enterprise search



QUERY → RETRIEVE → GENERATE

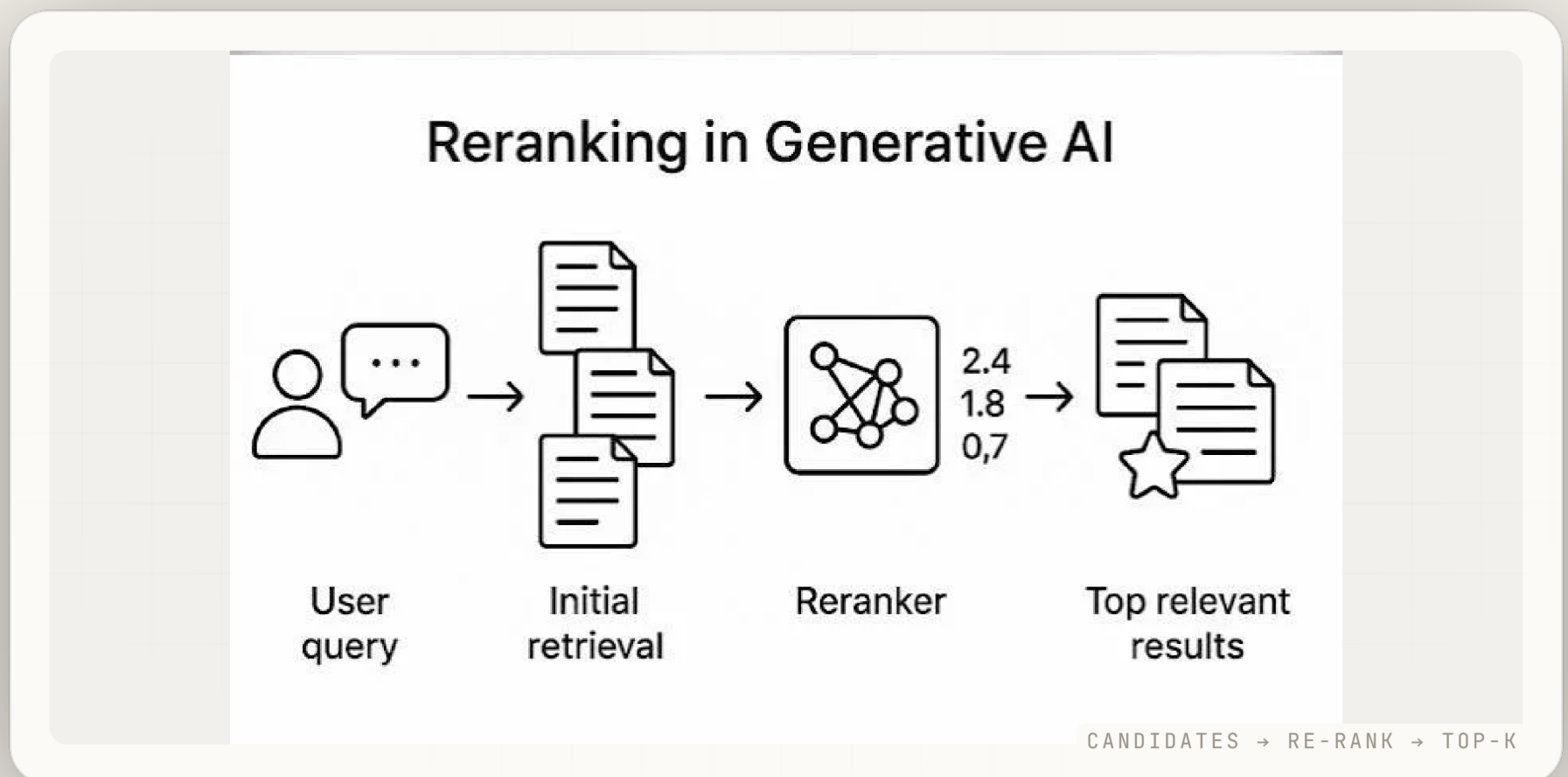
Reranker

re-score • reorder • surface the best

A model that **re-scores and reorders retrieved documents** to surface the most relevant ones, sharpening precision after a fast first-pass retrieval.

USED IN

- RAG pipelines
- Search
- Recommendations



ROTARY POSITION EMBEDDING • ARCHITECTURE

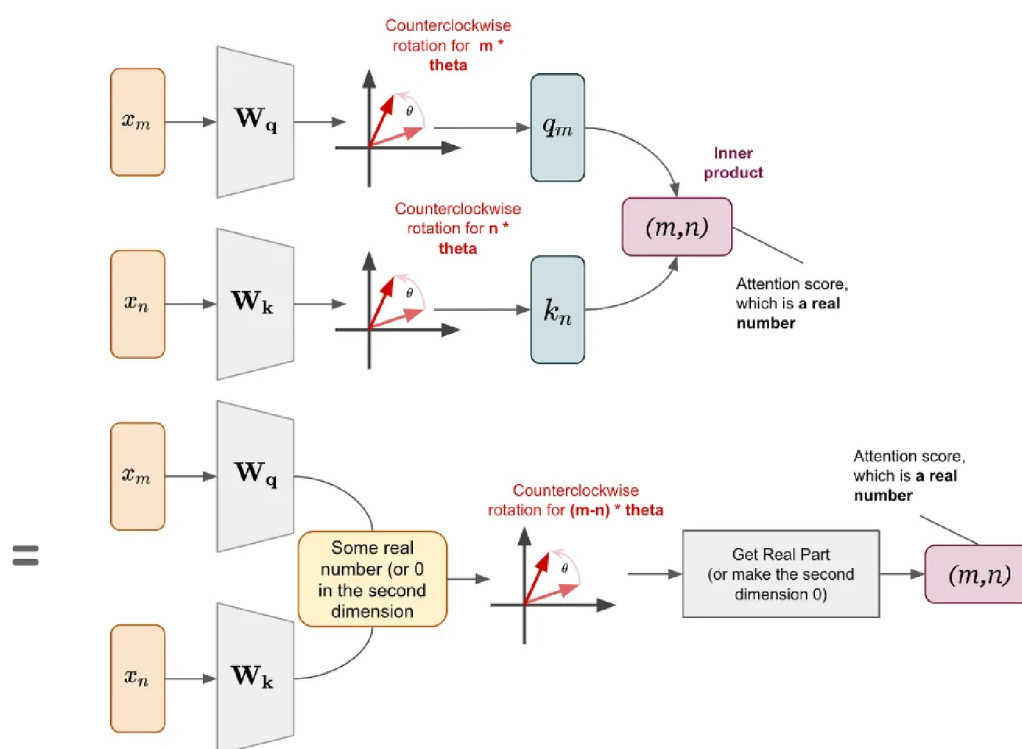
RoPE

rotate vectors • encode position

A way to encode token position by **rotating embedding vectors based on position**, enabling better long-sequence handling, used in LLaMA and most modern LLMs.

USED IN

- Transformers
- Long context
- LLaMA & GPT-NeoX



POSITION → ROTATION ANGLE

RAG ASSESSMENT • EVALUATION

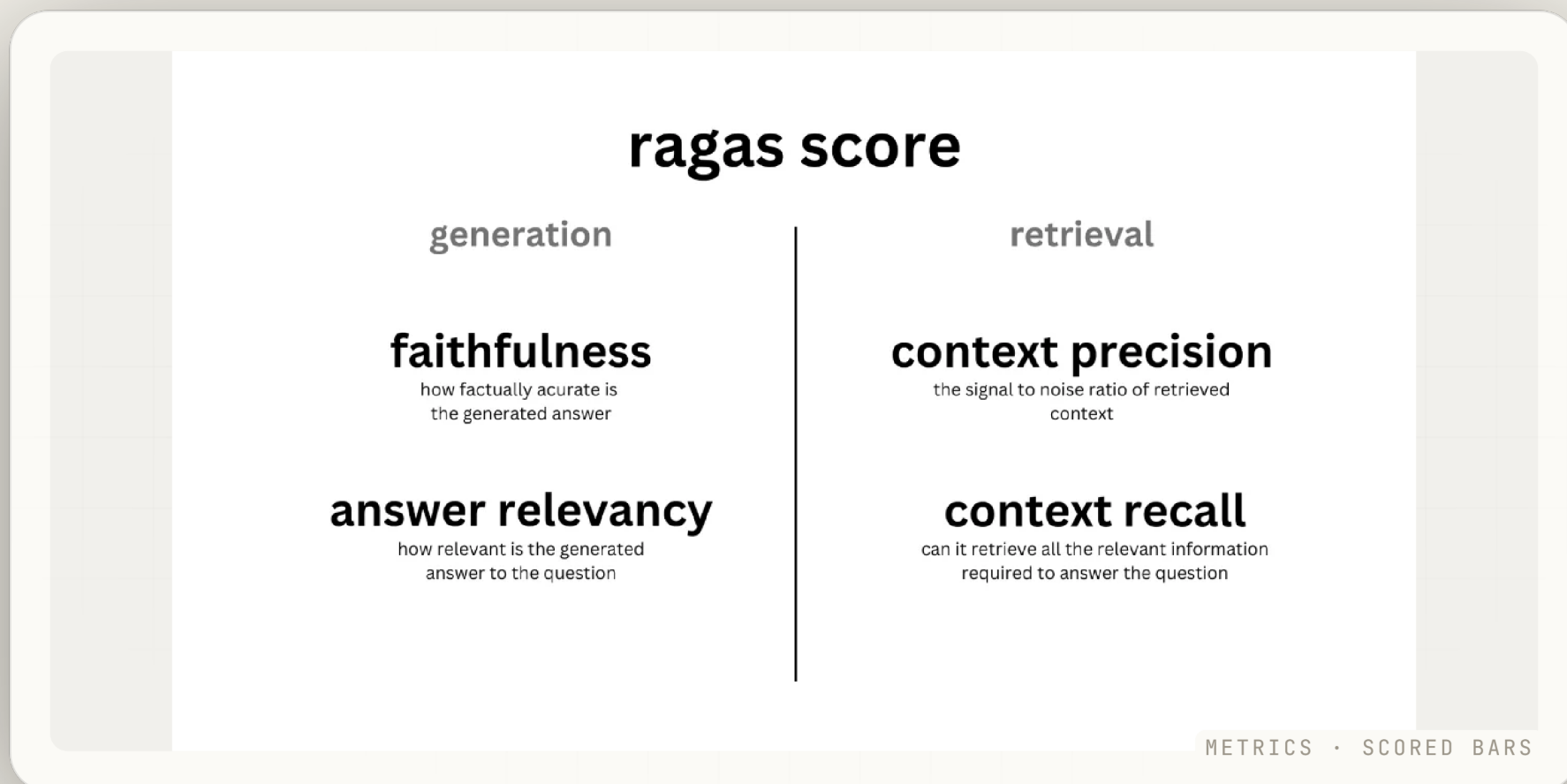
RAGAS

faithfulness • relevance • precision

An open-source framework for **evaluating RAG pipelines** on metrics like faithfulness, answer relevance, and context precision, benchmarking retrieval and generation quality.

USED IN

- RAG evaluation
- Benchmarking
- LLM ops



COEFFICIENT OF DETERMINATION • EVALUATION

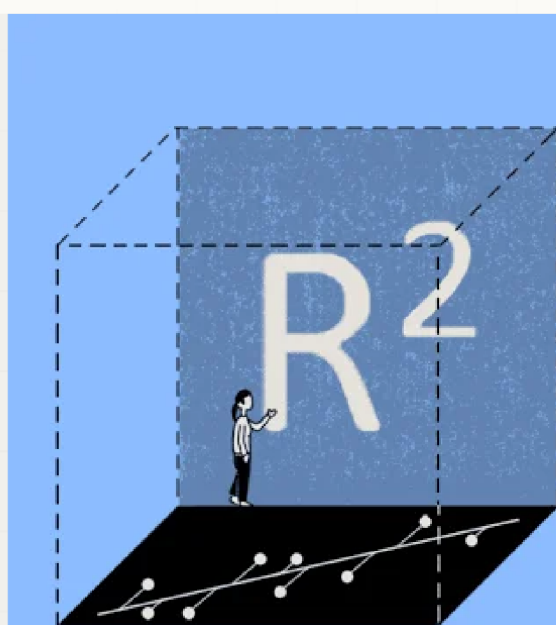
R^2 score

variance explained • 0 to 1

A regression metric showing **how much variance in the target the model explains**, ranging from 0 (no fit) to 1 (perfect fit).

USED IN

- Regression
- Model fit
- Forecasting



R-Squared

['är 'skwerd]

A statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model.

Investopedia

PREDICTED VS ACTUAL • FIT LINE

REPLICATION • RESEARCH PRACTICE

Replication

reproduce • verify • trust

Reproducing a research result using the original methodology, **a critical check on validity and reliability** of AI research findings.

USED IN

- Research rigor
- Benchmarks
- Peer review



Drop diagram · Replication

METHOD → RERUN → SAME RESULT

RELATIONAL DATABASE SYSTEM • INFRASTRUCTURE

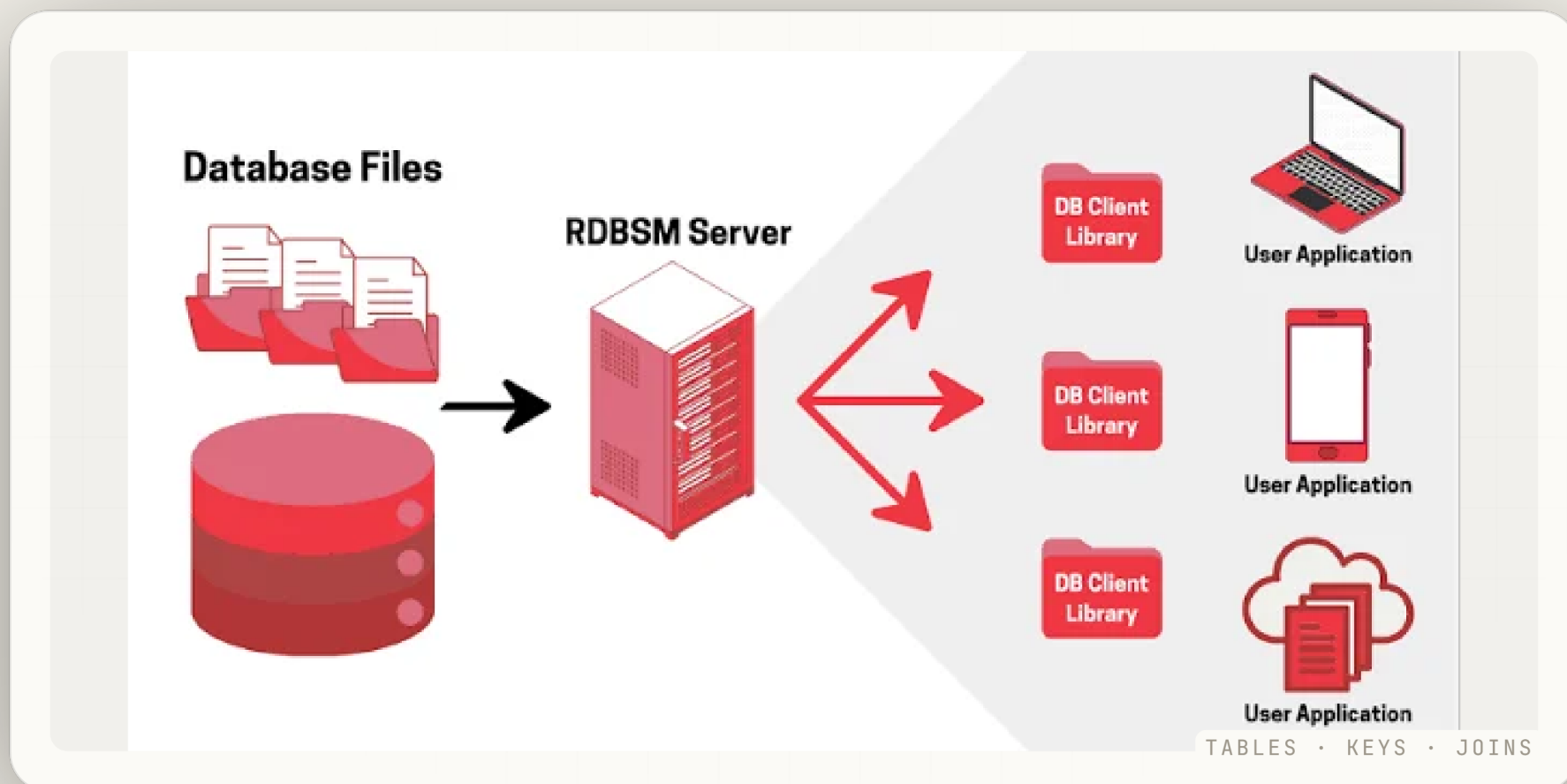
RDBMS

tables • rows • SQL • joins

A database organizing data into **structured tables queried via SQL**, set apart from NoSQL and vector stores by its rigid schema and join-based querying.

USED IN

- Transactional apps
- Analytics
- Feature stores



↓ UP NEXT • TOOLS & COMPANIES

The R stack.

Two R-named tools the AI stack quietly runs on, one **serves your data in milliseconds**, the other runs any open model behind one API.



IN-MEMORY DATA STORE

Redis



cache • vector search • sub-ms

The world's fastest in-memory database, now with **vector search and hybrid retrieval**, widely used for caching and RAG in production LLM apps.

In-mem

Sub-millisecond reads and writes

Vector

Hybrid search for RAG

2009

Created by Salvatore Sanfilippo

WHAT IT IS

In-memory key-value store and cache.

IN AI STACKS

Caching, sessions, and vector retrieval.

WHY DEVS USE IT

Fastest layer in front of any database.

FIND THEM

redis.io

MODEL HOSTING PLATFORM

Replicate

Replicate

run any model • one API

Model-agnostic infrastructure that lets you **run any open-source model via API**, image, video, language, audio. Think "Vercel for machine learning."

Any model

Image, video, audio, language

One API

Run models without managing GPUs

Open

Huge community model library

WHAT IT IS

API platform for running ML models.

IN AI STACKS

Inference for SD, Flux, LLMs, and more.

WHY DEVS USE IT

Ship model features without infra.

FIND THEM

replicate.com

↓ UP NEXT • PEOPLE TO FOLLOW

The R minds.

Three R-named researchers and educators **worth a follow**, from frontier AI safety to the daily firehose of what's new.

R

AI SAFETY RESEARCHER

Rohin Shah

Google DeepMind • AGI safety



Head of AGI Safety & Alignment at Google DeepMind and a **leading voice on technical AI safety**, writes the long-running Alignment Newsletter.

DeepMind

Head of AGI Safety & Alignment

Berkeley

PhD, Center for Human-Compatible AI

Newsletter

Long-running Alignment Newsletter

KNOWN FOR

Technical AI safety and alignment.

BEST FOR

Alignment and safety fundamentals.

WHY FOLLOW

Clear thinking on frontier-model risk.

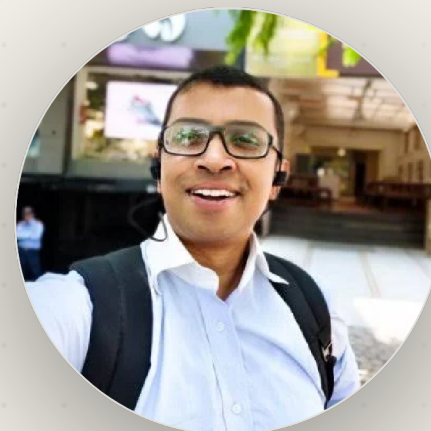
FIND THEM

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ML ENGINEER & CURATOR

Rohan Paul

X • @rohanpaul_ai



ML engineer who runs [the largest daily AI newsletter and show on X](#), plus a YouTube channel of ML, DL, and LLM fine-tuning tutorials.

Daily

Largest daily AI show on X

ex-DB

Former Deutsche Bank ML engineer

Tutorials

ML, DL, and LLM fine-tuning

KNOWN FOR

High-volume daily AI news on X.

BEST FOR

Staying on top of what's new.

WHY FOLLOW

Keep current without the doomscroll.

FIND THEM

x.com/rohanpaul_ai

AI EDUCATOR

Rahul Agarwal

LinkedIn • founder of MLWhiz



ML engineer and founder of MLWhiz, making machine learning accessible through **practical blogs, open-source, and hands-on tutorials.**

MLWhiz

Founder of the ML blog

Practical

Hands-on, applied tutorials

Open

Open-source projects & mentorship

KNOWN FOR

MLWhiz blog and applied ML guides.

BEST FOR

Going from theory to working code.

WHY FOLLOW

Approachable, practical ML teaching.

FIND THEM

linkedin.com/in/rahulagwl

The R stack, in one card.

Save it. Drop a comment with the R-term you use most.

01 RNN • ReLU

foundations

02 RL • RLHF

learning

03 Reasoning

inference

04 ReAct

agents

05 RAG • Reranker

retrieval

06 RoPE

architecture

07 RAGAS • R² score

eval

08 Replication

research

09 RDBMS

infra

+ Redis • Replicate

tools

@ Rohin • Rohan • Rahul

people



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

S Next